Homework 1
Introduction to Complex Analysis
(MATH 4230-40), Spring 2022

Due: 01/28/2022

Section 1.1 - Introduction to Complex Numbers

1. Express the following complex numbers in the form $a+bi$:

   (a) $(2+3i)(4+i)$

   (b) $(8+6i)^2$

   (c) $\left(1+\dfrac{3}{1+i}\right)^2$

2. Is it true that $\mathrm{Re}(zw) = (\mathrm{Re}z)(\mathrm{Re}w)$? Why or why not?

3. Simplify the following:

   (a) $(1+i)^4$

   (b) $(-i)^{-1}$

   (c) $\dfrac{1+i}{1-i}$

4. Simplify the following:

   (a) $\sqrt{1+\sqrt{i}}$

   (b) $\sqrt{1+i}$

   (c) $\sqrt{\sqrt{-i}}$

Section 1.2 - Properties of Complex Numbers

5. Solve the following equations:

   (a) $z^5 - 2 = 0$

   (b) $z^4 + i = 0$

6. What is the complex conjugate of $\dfrac{(3+8i)^4}{(1+i)^{10}}$?

7. Let $w \neq 1$ be an n^{th} root of unity. Show that $1 + w + w^2 + \ldots + w^{n-1} = 0$.

8. Assuming either $|z| = 1$ or $|w| = 1$ and $\overline{z}w \neq 1$, prove that

$$\left|\frac{z-w}{1-\overline{z}w}\right| = 1.$$